Pre-Analysis Plan

#ManyDaughters
Impact of Having Daughters on Attitudes, Preferences, and Behaviors

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1. Introduction

While gender differences in labor market outcomes have decreased over the last decades, considerable gender gaps in terms of income, labor supply and occupational choices remain (Blau & Kahn, 2017). Previous research shows that the birth of the first child is crucial in explaining the divergence in male and female outcomes (Cortés & Pan, 2023; Kleven et al., 2019). Several scholars find that gender norms are important in explaining the persistence of these gender gaps and show transmission of those norms to the next generations (Alesina et al., 2013; Boelmann et al., 2025).

Less is known about the impact that children have on their parents' norms and values. However, there is evidence that the gender of the child(ren) matters for parental outcomes. Different studies show that the impact of having daughters may be manyfold: Borrell-Porta et al. (2019) show that rearing daughters has a negative effect on holding traditional gender norms. Within firms, the income of women increases when managers have a daughter compared to having a son (Ronchi & Smith, 2021). Bhalotra et al. (2025) show that the child penalty in earnings and in mental health is bigger for mothers of daughters than mothers of sons. Judges who have a daughter display more feminist voting behavior (Glynn & Sen, 2015). Having a daughter affects political preferences, voting behavior or attitudes towards abortion right (Greenlee et al., 2020; Sharrow et al., 2018; Van Effenterre, 2020). Firms whose chief executive officer has a daughter achieve higher ratings in corporate social responsibility (Cronqvist & Yu, 2017). Given that the effect of children on their parents seems to vary by context, it is crucial to better understand the magnitude and the robustness of these effects.

This PAP forms part of the #Manydaughters project, that invited researchers to analyze the effect of having daughters on attitudes, preferences and behaviors[1]. The outcomes of interest are gender norms in terms of attitudes, gender equality in terms of behavior and political preferences. We follow the project instructions and estimate only one coefficient for each of the four given hypotheses. We write our pre-analysis plan based on a five percent sample of the Socio Economic Panel[2]. The analysis plan is thus not to be equated with the ideal analysis as we do not believe that the Socio Economic Panel is the ideal dataset for this analysis or that these hypotheses can be tested using a single coefficient. The effect of daughters might vary over time, but e.g. event study analyses are not possible given the project restrictions. The effect of daughters might also vary along with the outcomes and thus a single (average) estimate might be misleading.

[1] For further information, see the project website of the organizers.
[2] Given that this is not a publicly available version of the SOEP, we can unfortunately not cite it.

Hypotheses to be tested:

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes)*

H2. *Having daughters leads to more left-leaning political preferences*

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)*

The decision to have none, one, or several children is potentially endogenous and might be correlated with each of the outcomes that we want to analyze. For example, parents with more egalitarian attitudes and behaviors might have a higher probability of being both employed and supplying a similar number of hours worked at the market. Due to time constraints, they might opt for a smaller number of children than a couple that follows the traditional male-breadwinner model. In the past, societies that hold more traditional norms have favored male heirs, resulting in a preference for sons (Bhalotra et al., 2020; Choi & Hwang, 2015; Dahl & Moretti, 2008; Glogowsky et al., 2023). For our identification strategy, we therefore want to exploit the sex of the first-born child as a natural experiment. We assume that the gender of the first child is as good as random and estimate the effect of having a daughter, versus having a son, as a first child, on the outcomes.

2. The effect of daughters on support for gender equality

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes)

2.1. Variables

The SOEP contains individual (*pid*), household (*hid*) and survey wave and month (*syear*, *pgmonth*) identifiers that we will use throughout our analysis. We further use information from the parents' birth biography questionnaire to identify the timing (year and month) of childbirth and the reported sex of the child. For each hypothesis, we will use additional variables for the outcomes and generate variables according to our identification strategy

Variable: **plh0308_v2**

We measure our main outcome for H1, support for gender equality in terms of attitudes, based on a question that was included in 2018. Individuals were asked to state agreement or disagreement with the following statement: "*The best situation is when both the man and the woman are equally employed and both take equal responsibility for the household and the family*." Respondents can rate their answer on a scale between 1 (strongly disagree) to 7 (strongly agree).

Variable: **d_daughter**

(kidsex01)

We use information on the sex of the first-born child to create our main explanatory variable, an indicator equaling one when the first child is female.

Variables: **n_brothers,** **n_sisters**

(syear, pgmonth, kidgeb01 – kidgeb18, kidmon01 – kidmon18, kidsex01 – kidsex18)

We use variables on the sex and timing of birth of each child to determine the number of already born brothers and sisters of a first child at the time of the survey wave.

Variable: **fchild_age**

(pgmonth, kidgeb01, kidmon01)

We use the survey month and the year of month of first childbirth to determine the child's age in years in the respective survey wave.

2.2. Inclusion / exclusion criteria

We limit our sample to parents who we observe after their first child was born and who supply information on their children's birth biography. We further keep only those individuals for whom our proxy for gender norms is filled, which limits our observation period to survey year 2018.

2.3. Analysis

In general, individuals might select into parenthood based on characteristics that correlate with gender norms. We want to exploit that the sex of the first child randomly treats individuals with either having a daughter or having a son. We thereby compare parents who have a daughter as a first child to parents who have a son as a first child.

Equation 1 shows our estimation strategy for H1:

$$Y_i = \alpha_1 d_daughter_i + \alpha_2 X_i + \varepsilon_i \quad (1)$$

We estimate the effect of having a daughter compared to having a son as a first child on gender norms. Y_i represents our outcome variable. d_daughter represents the indicator variable for having a daughter as a first child. Our coefficient of interest is estimated by α_1. X_i, represents a set

of control variables: To account for the fact that having any additional child or a daughter as a subsequent child might also affect gender norms, we control for the number of brothers and for the number of sisters of the first-born child at the time of the survey.

Our approach relies on the assumption that the sex of the first child is exogenous to parents. The assumption would be violated if parents had an influence on the sex of their first born, for example through selective abortion of unborn girls or boys. §15 of the Genetic Diagnostics Act is designed to prevent parents from aborting an unborn child based on undesired sex. Until the 14th week of pregnancy, abortions are generally illegal under criminal law, but they are not prosecuted and remain exempt from punishment under certain conditions. After the 14th week of pregnancy has passed, abortions are only permitted if the pregnant woman faces a life-threatening situation or the risk of a serious impairment to her physical or mental health. We therefore think that our assumption holds for the German context. By being constrained to one coefficient we cannot provide balance tables that would provide more faith in this assumption. We requested clarification from the #Manydaughters team but were only referred to the PAP guidelines which state the one-coefficient goal.

We take the family as the unit that is treated and hence based on Abadie et al. (2022)cluster standard errors at the household level.

2.4. Results

We interpret a positive and significant α_1 such that having a daughter as a first child increases the probability to support gender equality, which would be in line with H1. Consistently, we interpret a negative α_1 such that having a daughter is negatively correlated with support for gender equality. If α_1 is significant and close to zero, we consider daughters not to have any effect on parental gender norms.

3. H2: The effect of having daughters on political preferences

H2. Having daughters leads to more left-leaning political preferences

3.1. Variables

We use to the same set of variables as depicted in Section 2, except for the main outcome variable.

Variable: **left** *(plh0004)*

In the years 2005, 2009, 2014 and 2019, respondents are asked "*Where would you place your own political views?*". They can choose a value between zero to ten, where zero corresponds with "*far left*" and ten corresponds with "*far right*". We recode the variable such that a value of ten corresponds to far left and a value of zero corresponds to far-right political tendency, such that moving up the scale is equivalent to moving towards more left-leaning political preferences.

3.2. Inclusion/exclusion criteria

We apply the same restrictions as outlined in Section 2.2. We limit our sample to parents with information on their children's birth biography and keep those observations that contain information about political preferences. Therefore, our observation period is limited to survey years 2005, 2009, 2014 and 2019.

3.3. Analysis

Equation 2 shows our estimation strategy for H2:

$$Y_{i,t} = \alpha_1 d_daughter_i + \alpha_2 X_{i,t} + \delta_t + \varepsilon_{i,t} \quad (2)$$

We estimate the effect of having a daughter compared to having a son as a first child on political preferences. $Y_{i,t}$ represents our outcome variable, a value for political preferences between far right and far left, ranging from zero to ten. As before, variable d_daughter represents the indicator variable for having a daughter as a first child and our coefficient of interest is estimated by α_1. The rest of the equation remains unchanged, $X_{i,t}$ represents the same set of control variables, and we include survey wave fixed effects, represented by δ_t to capture any influence common to all parents in that specific wave —like macro shocks, policy changes, or trends—thereby controlling for time-specific confounders.

We take the family as the unit that is treated and hence based on Abadie et al. (2022)cluster standard errors at the household level.

3.4. Results

If α_1 is positive and significant, we conclude that having a daughter as a first child is associated with more left-leaning political attitudes (in accordance with H2). Consistently, we interpret a negative α_1 such that having a daughter as a first child is associated with less left-leaning political attitudes. If α_1 is significant and equal or very close to zero, we consider daughters not to have any effect on political preferences.

4. H3: The effect of daughters on household equality

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

4.1. Variables

Variable: **childcare** *(pli0044_h)*

Behaviors or outcomes that reflect gender equality in the household are manyfold. As outlined, having a child is a decisive event in explaining gender gaps. We therefore focus on the amount of time spent by parents on childcare on a usual working day (weekdays), measured in hours, to create our main outcome variable. We combine information on a (heterosexual) couples' time spent on childcare and take the absolute value of the difference as a measure of household gender equality:

$$childcare_diff_h = |\ hours_{i,h} - \ hours_{i-1,h}| \quad (3)$$

A value of zero implies that both partners spend an equal amount of time on childcare, any divergence from zero indicates that one of the partners spends more time on childcare than the other.

4.2. Inclusion/exclusion criteria

Partner indicators (parid from ppathl and **pgpartnr** from pgen) have the purpose of defining couples in SOEP households and thus to make possible analyses on the dyadic level. Persons without spouse and (cohabitating) partner receive a missing code "-2" (=does not apply).

We first keep all men in the sample and then join information on female partners that we can identify via the partner id (pgpartnr) in the SOEP. The partner ids mark couples in SOEP households. We drop all individuals with missing partner and where information on weekday hours of childcare is missing, to create our household measure of gender inequality. Questions on hours spent on childcare on weekdays are part of the SOEP from 1985 onwards. We can therefore use all waves of the SOEP to analyze H3. We further limit our sample to all individuals where we have information on childbirth (sex and birth year). We exclude individuals who are surveyed in the same month as the birth of the first child as we don't know whether the child is already born.

4.3. Analysis

We use a simple difference-in-differences setup, where we compare households who had a girl to parents who had a boy before and after giving birth to their first child. We limit our sample to individuals surveyed within 12 months before and 12 months after giving birth to the first child because we want to separate the effect of having a daughter from the potential effects of additional children. Our main identification assumption requires that, in the absence of the treatment (i.e., having a girl), parents who had daughters and those who had sons would have shown similar trends in the outcome variable prior to birth. For our baseline specification, we estimate

$$Y_{h,t} = \alpha_1 treated_h + \alpha_2 post_t + \alpha_3 interaction_{t,h} + \ \varepsilon_{h,t} \quad (4)$$

$Y_{h,t}$ represents our outcome variable. The main difference to previous equations is that now we are estimating on the household, as a proxy for family, level. The coefficient of interest is α_3 which is the difference-in-differences coefficient. We take the family (household) as the unit that is treated and hence based on Abadie et al. (2022) cluster standard errors at the household level.

4.4. Results

If α_3 is positive and significant, we conclude that having a daughter as a first child is associated with more household equality (in accordance with H3). Consistently, we interpret a negative α_3 such that having a daughter as a first child is associated with less household equality. If α_3 is significant and equal or very close to zero or insignificant, we consider daughters not to have any effect on household equality.

5. H4: The effect of daughters on gender norms by gender

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)

5.1. Variables

We generate the same set of variables as outlined under H1 but additionally use a dummy variable that indicates whether the parent is of male / female gender. Inclusion/Exclusion criteria

5.2. Analysis

We build on the same estimation strategy as outlined under H1, but we allow our specification to vary by parental sex.

$$Y_{i,t} = \alpha_1 d_daughter_i * father + \alpha_2 d_daughter_i + \alpha_3 father_i + \alpha_4 X_{i,t} + \delta_t + \varepsilon_{i,t} \quad (5)$$

We estimate the effect of having a daughter compared to having a son as a first child on gender norms by gender of the parent. Y_i represents our outcome variable. d_daughter represents the indicator variable for having a daughter as a first child. father represents the indicator variable for being a father. Our coefficient of interest is estimated by α_1. X_i, represents a set of control variables: To account for the fact that having any additional child or a daughter as a subsequent child might also affect gender norms, we control for the number of brothers and for the number of sisters of the first-born child at the time of the survey.

Our approach relies on the assumption that the sex of the first child is exogenous to parents. The assumption would be violated if parents had an influence on the sex of their first born, for example through selective abortion of unborn girls or boys. §15 of the Genetic Diagnostics Act is designed to prevent parents from aborting an unborn child based on undesired sex. Until the 14th week of pregnancy, abortions are generally illegal under criminal law, but they are not prosecuted and remain exempt from punishment under certain conditions. After the 14th week of pregnancy has passed, abortions are only permitted if the pregnant woman faces a life-threatening situation or the risk of a serious impairment to her physical or mental health. We therefore think that our assumption holds for the German context.

We take the family as the unit that is treated and hence based on Abadie et al. (2022)cluster standard errors at the household level.

5.3. Results

We interpret a positive and significant α_1 such that having a daughter as a first child increases the support gender equality for men more than for women, which would be in line with H4. If α_1 is significant and close to zero, we consider daughters not to have any differential effect on women and men.

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